Fiscal Year:	FY 2018	**Task Last Updated:**	FY 08/30/2018
PI Name:	Sibonga, Jean Ph.D.		
Project Title:	Feasibility Study: QCT Modality for Risk Surveillance of Bone - Effects of In-flight Countermeasures on Sub-regions of the Hip Bone		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Bone Fracture**:Risk of Bone Fracture due to Spaceflight-induced Changes to Bone (2) **Osteo**:Risk Of Early Onset Osteoporosis Due To Spaceflight		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jean.sibonga-1@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	281-483-4556
Organization Name:	NASA Johnson Space Center		
PI Address 1:	Bone Mineral Laboratory, SK 311		
PI Address 2:	2101 NASA Parkway		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	Directed Research
Start Date:	12/01/2011	**End Date:**	09/30/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
Contact Email:	Peter.norsk@nasa.gov		
Flight Program:	Pre/Post Flight		
Flight Assignment:	ISS NOTE: End date changed to 9/30/2018 per PI (Ed., 9/10/18) NOTE: Changed end date to 7/31/2016, although this is tentative, per PI saying work has not yet been completed (Ed., 4/29/16) NOTE: Gap changes per IRP Rev E (Ed., 1/27/14) NOTE: Title change per HRP and PI to "Feasibility Study: QCT Modality for Risk Surveillance of Bone - Effects of In-flight Countermeasures on Sub-regions of the Hip Bone"; previously "Occupational Risk Surveillance for Bone: Pilot Study - Effects of In-flight Countermeasures on Sub-regions of the Hip Bone" (Ed., 1/23/2013)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):			
Grant/Contract No.:	Directed Research		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	Measurement of areal bone mineral density [aBMD, g/cm2] by dual-energy x-ray absorptiometry [DXA] is required by NASA for assessing skeletal integrity in astronauts. Advantages of DXA include the facts that aBMD is widely-applied predictor of fractures in the aging population and that there are aBMD-based guidelines for identifying persons at high risk for osteoporotic fractures. In contrast to the 2-d imaging by DXA, quantitative computed tomography [QCT] is a 3-d bone imaging technology that is used typically to scan the hip and spine. QCT is capable of measuring, volumetric BMD [BMD, mg/cm3] of separate cortical and trabecular sub-regions as well as of total (integral) bone. In contrast to the 2-d imaging by DXA, volumetric QCT at the hip is limited to research applications at this time because there is not enough medical evidence to determine how QCT data should be used in clinical practice. QCT however provides additional information on bone structure and increases the understanding of how bones respond to effectors of bone loss or gain. NASA recently convened a panel of clinical bone experts to review available medical and research information from astronauts who flew on long-duration space missions. As part of its charge, the panel identified a clinical trigger upon which the flight surgeon should have the astronaut evaluated further by an osteoporosis specialist. Specifically, the Panel recommended that if restoration to preflight BMD is not observed for the hip trabecular compartment at two years after return to Earth, then that astronaut should be evaluated for possible therapeutic intervention to prevent premature osteoporotic fractures. This pilot study proposes to use preflight and postflight QCT scanning of the hips in International Space Station (ISS) astronauts to evaluate the ability of in-flight countermeasures to prevent the occurrence of this clinical trigger. This study further hypothesizes that QCT scanning can distinguish the effects of different categories of in-flight countermeasures/activities on distinct sub-regions of the hip bone. For example, this pilot study will demonstrate that biochemically-based countermeasures (e.g., dietary manipulation of acidic to basic amino acid intake or bisphosphonates medication) will have a detectable prevention of BMD loss in hip trabecular compartment while biomechanically-based countermeasures (exercise regimens) will have detectable expansion of cortical bone apposition -- increasing both bone cross-sectional area and integral BMD as a consequence. These different effects on hip morphology will be subsequently translated to an effect on hip bone strength of the ISS astronaut. The combination of countermeasures that impact both compartments will more likely result in greater hip bone strength -- as estimated by analyzing QCT data by Finite Element Modeling (FEM) -- than of any singly applied countermeasure. This assertion will be approached in this pilot study by addressing the following Aims in each ISS astronaut: 1) Characterize the response of i) trabecular and cortical BMDs of the hip and ii) cross-sectional areas of cortical bone, trabecular bone, and integral bone, to countermeasures that are either based upon biochemistry or mechanical-loading -- with QCT measures. 2) Translate the QCT-measured changes in hip bone morphology (Aim 1) to hip bone fracture loads (aka, "hip bone strength") using FEM. 3) Characterize QCT-measured changes in hip bone morphology (Aim 1) following a 12-month postflight period on Earth and, in addition, translate these changes to the percentage recovery of preflight hip bone strength determined by FEM. By addressing these aims this pilot study, using a research tool, will provide preliminary data that are critical for clinical issues related to fracture risk: Are in-flight countermeasures and postflight activities sufficient to protect against incidence of a clinical trigger for medical intervention? Do countermeasures protect against a decline in bone strength? Can hip bone strength be sufficiently recovered? In addition, Trabecular Bone Score (TBS) analysis of DXA lumbar spine scans will be used to characterize bone microarchitecture of the lumbar spine and to determine if an effect of space flight can be detected in the retrospective ISS DXA data set. Like QCT, TBS may help fill a void with traditional DXA measurements, as it can differentiate between areas of bone that have the same areal BMD value but different 3-dimensional microsarchitecture in the trabecular bone compartment of the spine. This is achieved by retrospectively analyzing areal DXA scans and measuring the mean rate of variation of gray levels. Since QCT measurements in this study are being obtained only on the hip, TBS analyses will serve a similar purpose for the spine. In addition to the TBS analysis of ISS DXA scans, TBS analyses of retrospective precision study subjects will be used to determine measurement precision of this technique on a population similar to that of the astronauts.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research, which requires focused and constrained data gathering and analysis that is more appropriately obtained through a non-competitive proposal.
Research Impact/Earth Benefits:	Research Impact: This study will provide data in addition to the medically-required measurement of aBMD by DXA. There is a requirement in the osteoporosis field to expand evaluations beyond DXA aBMD (i.e., "Bone Quality") to evaluate fracture risk fully because aBMD does not account for 100% of bone strength. This requirement is particularly important for the subject with poorly defined bone loss, i.e., other than age-related bone loss. Moreover, a report of preflight and postflight QCT data from eleven ISS astronaut reveals that changes in hip bone strength by FEM do not correlate with changes in DXA aBMD. This absence of correlation suggests that DXA aBMD does not detect all of the changes in bone strength due to spaceflight that can be detected by QCT and FE modeling. Earth Benefits: This expanded assessment of skeletal integrity, being validated for space flight-induced bone loss in astronauts, would be relevant for the terrestrial, complicated patient (e.g., glucocorticoid-induced, alcohol-induced). Recently, FEM estimations of bone strength have been evaluated in population studies as predictors of incident hip fractures. These FE hip strengths are being evaluated for cut-points that would provide thresholds of acceptable bone health for active astronauts and aging retired astronauts. The development of these cut-points, as demonstrated for astronauts, would undergird the current discussions to use FE hip strength as a substitute for expensive and time-consuming prospective trials with fracture outcome -- the standard validation of hip fracture interventions.

Task Progress:	The original intent of this study (“Hip QCT”) was to conduct a pilot study of 10 long-duration crewmembers who were already participating in other research studies involving pre and post-flight QCT scans of the hip. These subjects had already participated, or were scheduled to participate, in QCT testing as part of the Bisphosphonate SMO (supplemntary medical objective) or Sprint flight studies: preflight, R+1 week and R+1 year scans in the case of the SMO; and preflight and R+1 week scans in the case of the Sprint study. Sprint study subjects who consented to participate in the Hip QCT study were asked to participate in an additional scan at R+1 yr. All 10 subjects agreed to participate in a final, additional scan at R+2 years, but only if R+1-year testing indicated incomplete recovery in bone mineral density (BMD) of the trabecular compartment of the hip. Of the 10 subjects, 5 met the R+1-year criterion for further testing, and all 5 participated in R+2-year scans. In addition to the 10 original subjects, the study was allowed to include an 11th subject—one crewmember of the 1-year ISS flight—who would participate in the same QCT testing schedule (preflight, R+1 week, R+1 year, and R+2-year if recovery at 1 year was incomplete). In addition to the QCT scans, bone densitometry data by dual energy X-ray absorptiometry (DXA) and bone-related biochemistry data were requested on all subjects and obtained via data-sharing with other studies or medical requirements. All scheduled testing sessions for the 10 original subjects and the 1-year mission subject were completed by March 2018. QCT scans were submitted to Dr. T. Lang at UCSF (University of California San Francisco) for analysis, including measures of cortical and trabecular BMD, as well as estimates of bone strength computed from Finite Element modeling of the QCT scans. The resulting data have been reduced and analyzed, and preliminary results from the n=10 group were presented at the NASA Human Research Program Investigators’ Workshop in Galveston, TX, (January 2018). Final results from the n=10 group are now being written up in a manuscript to be submitted for publication in a peer-reviewed scientific journal. Data from the 1-year-mission subject are excluded from this manuscript due to the confounding effects of longer space flight duration, but may be written up as part of a larger, multidisciplinary, effort to document results from the 1-year mission. There are two manuscript deliverables from the data generated by this study. i) “Supplemental Use of Quantitative Computed Tomography Hip Scans of Astronauts for Clinical Assessments at NASA” to be submitted to journal Aerospace Med Human Performance (Authors: J.D. Sibonga , E.R. Spector , B.E. Lewandowski , S.R. Zwart , S.M. Smith , T.F. Lang) and ii) “The Utility of Hip Scans by Quantitative Computed Tomography (QCT) to Assess the Mitigation of Skeletal Changes during Spaceflights” to be submitted to journal Bone. (Authors: J.D. Sibonga , E.R. Spector , A.D. Leblanc , B. Lewandowski , M.E. Downs , L. Ploutz - Snyder , T. Lang).
Bibliography Type:	Description: (Last Updated: 05/24/2021)
Articles in Peer-reviewed Journals	Sibonga JD, Spector ER, Johnston SL, Tarver WJ. "Evaluating bone loss in ISS astronauts." Aerosp Med Hum Perform. 2015 Dec;86(12 Suppl):A38-A44. https://doi.org/10.3357/AMHP.EC06.2015 ; PubMed PMID: 26630194 , Dec-2015
Articles in Peer-reviewed Journals	Sibonga JD, Spector ER, Keyak JH, Zwart SR, Smith SM, Lang TF. "Use of quantitative computed tomography to assess for clinically-relevant skeletal effects of prolonged spaceflight on astronaut hips." J Clin Densitom. 2020 Apr-Jun;23(2):155-64. Epub 2019 Aug 26. https://doi.org/10.1016/j.jocd.2019.08.005 ; PubMed PMID: 31558405 , May-2020